

파생금융상품 II: 옵션계약(Options contract)

Options contract

An options contract offers the buyer the opportunity to buy or sell the chosen underlying asset at a price set out in the contract either within a certain timeframe or at the expiration date.

right() -> premium

- **Buying an option offers the right, but not the obligation**, to purchase or sell the underlying asset.
- Can be powerful tools for hedging and speculation.

option holder(buyer) : right, limit loss (premium = sunk cost) option writer(se

참고자료: <https://www.cmegroup.com/ko/education/learn-about-trading/courses/introduction-to-options.html>

Call option

- A **call option** gives its holder **the right to buy** an asset:
 - At the **exercise or strike** price
 - On or before the expiration date

: before : on
- Exercise the option to buy the underlying asset if **market value > strike price.**

Put option

- A **put** option gives its holder **the right to sell** an asset:
 - At the exercise or strike price
 - On or before the expiration date
- Exercise the option to sell the underlying asset if market value $<$ strike price.

The Option Contract

- The purchase **price of the option** is called the **premium.**
- Sellers (writers) of options receive premium income.
- If holder exercises the option, the writer must make (call) or take (put) delivery of the underlying asset.

Profit and Loss on a Call: Example 1

- A January 2010 call on IBM with an exercise price of \$130 was selling on December 2, 2009, for \$2.18 (premium).
- The option expires on the third Friday of the month, or January 15, 2010.
- If IBM remains below \$130, the call will expire worthless.

Profit and Loss on a Call: Example 2

- Suppose IBM sells for \$197 on the expiration date.

- Option value = stock price - exercise price

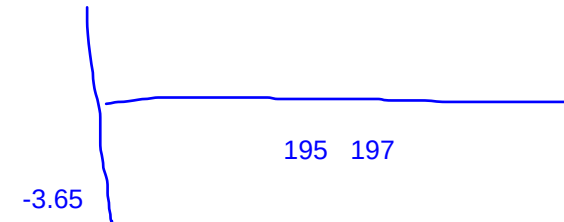
$$\$197 - \$195 = \$2$$

- Profit

= Final (option) value – Original investment (premium)

$$= \$2.00 - \$3.65 = -\$1.65$$

- Option will be **exercised to offset loss of premium**.
- Call will not be strictly profitable unless IBM's price exceeds \$198.65 (strike + premium) by expiration.



Profit and Loss on a Put: Example 3

- Consider a February 2013 put on IBM with an exercise price of \$195, selling on January 18, 2013 for \$5.00 (premium).
- Option holder can sell a share of IBM for \$195 at any time until February 15.
- If IBM goes above \$195, the put is worthless.

x

Profit and Loss on a Put

- Suppose IBM's price at expiration is \$188.
- Value at expiration = exercise price – stock price:
 $\$195 - \$188 = \$7$
- Investor's profit:
 $\$7.00 - \$5.00 = \$2.00$
- Holding period rate of return = 40% over 28 days!

Market and Exercise Price Relationships

In the Money - exercise of the option produces a positive cash flow

Call: exercise price $<$ market price

Put: exercise price $>$ market price

Out of the Money - exercise of the option would not be profitable

Call: market price $<$ exercise price.

Put: market price $>$ exercise price.

At the Money - exercise price and market price are equal

American vs. European Options

American - the option **can be exercised** at any time **before** expiration or maturity

$C = C_0$

European - the option can **only be exercised** on the expiration or maturity date

* The majority of CME Group options on futures are European style and can be exercised only at expiration. Some of the notable exceptions that have **American style** expiration are the quarterly options on the **S&P500 futures contracts, Eurodollar options, and Treasury options.**

Different Types of Options

- Stock Options
- Index Options
- Futures Options
- Foreign Currency Options
- Interest Rate Options

Payoffs and Profits at Expiration - Calls

Notation

Stock Price = S_T Exercise Price = X

Payoff to Call Holder: option value

$(S_T - X)$ if $S_T > X$

0 if $S_T \leq X$

-> > ; -> X

Profit to Call Holder

Payoff - Premium

Payoffs and Profits at Expiration - Calls

Payoff to Call Writer

$$\begin{aligned} &-(S_T - X) \quad \text{if } S_T > X \\ &0 \quad \quad \text{if } S_T \leq X \end{aligned}$$

Profit to Call Writer

Payoff + Premium

Payoff and Profit to Call Holders at Expiration

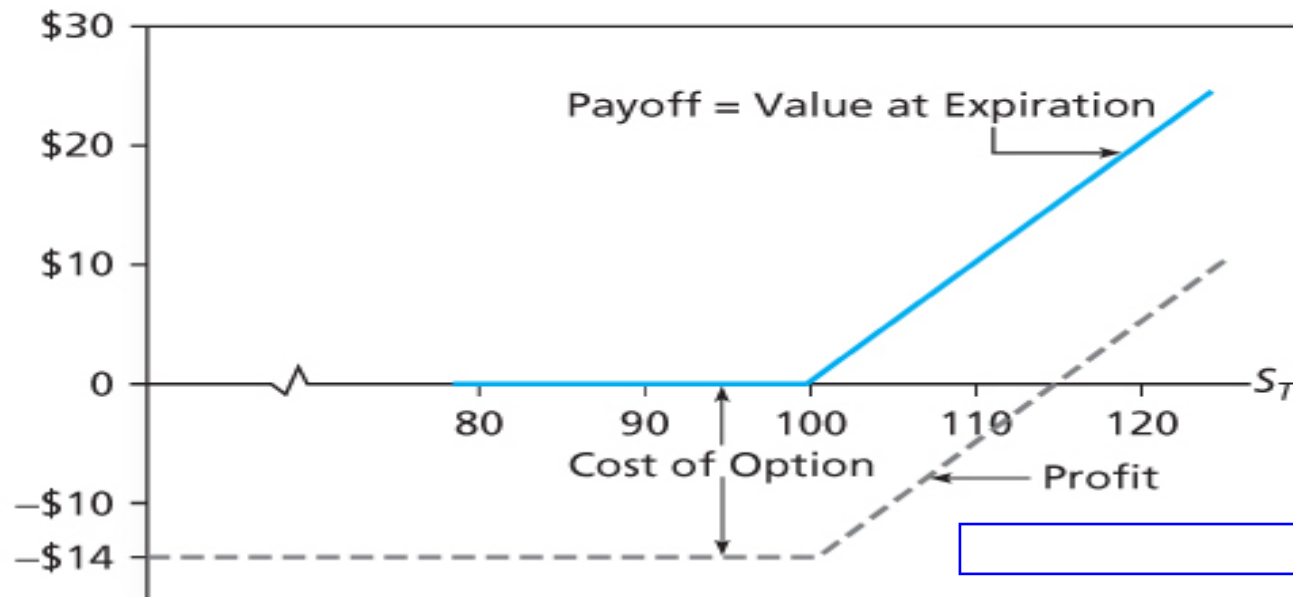


Figure 20.2 Payoff and profit to call option at expiration

Payoff and Profit to Call Writers at Expiration

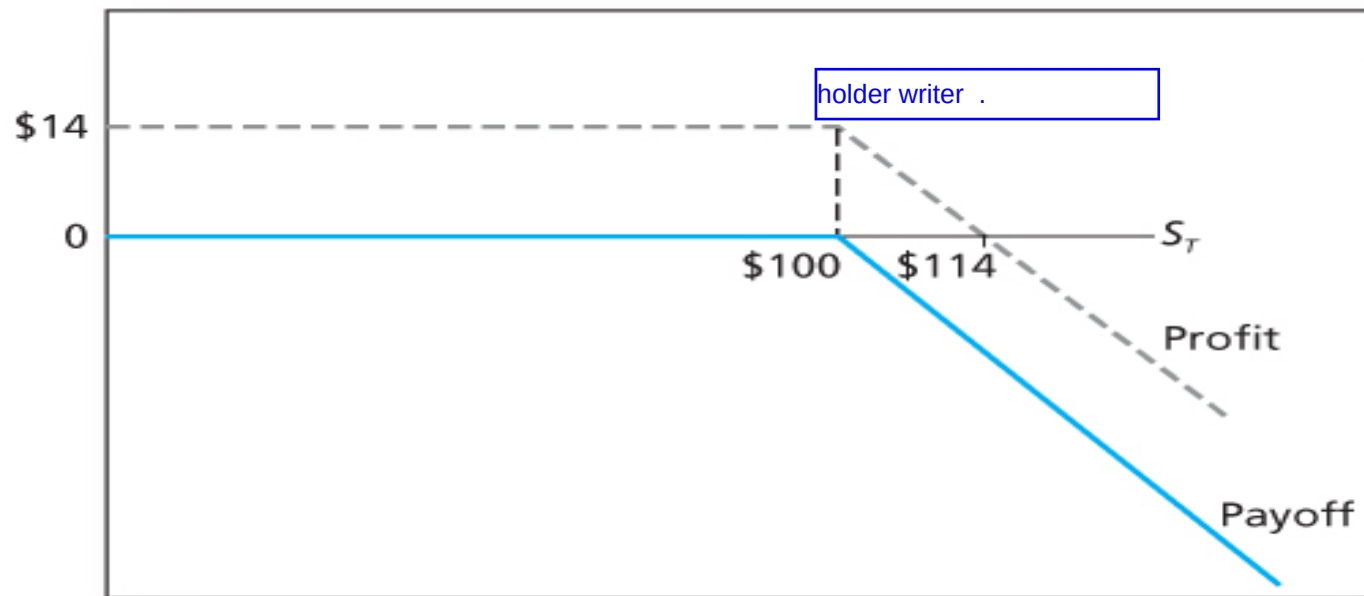


Figure 20.3 Payoff and profit to call writers at expiration

Payoffs and Profits at Expiration - Puts

Payoff to Put Holder



0 if $S_T \geq X$

$(X - S_T)$ if $S_T < X$

Profit to Put Holder

Payoff - Premium

Payoffs and Profits at Expiration – Puts

Payoff to Put Writer

$$0 \quad \text{if } S_T \geq X$$

$$-(X - S_T) \quad \text{if } S_T < X$$

Profit to Put Writer

Payoff + Premium

Payoff and Profit to Put Holders at Expiration

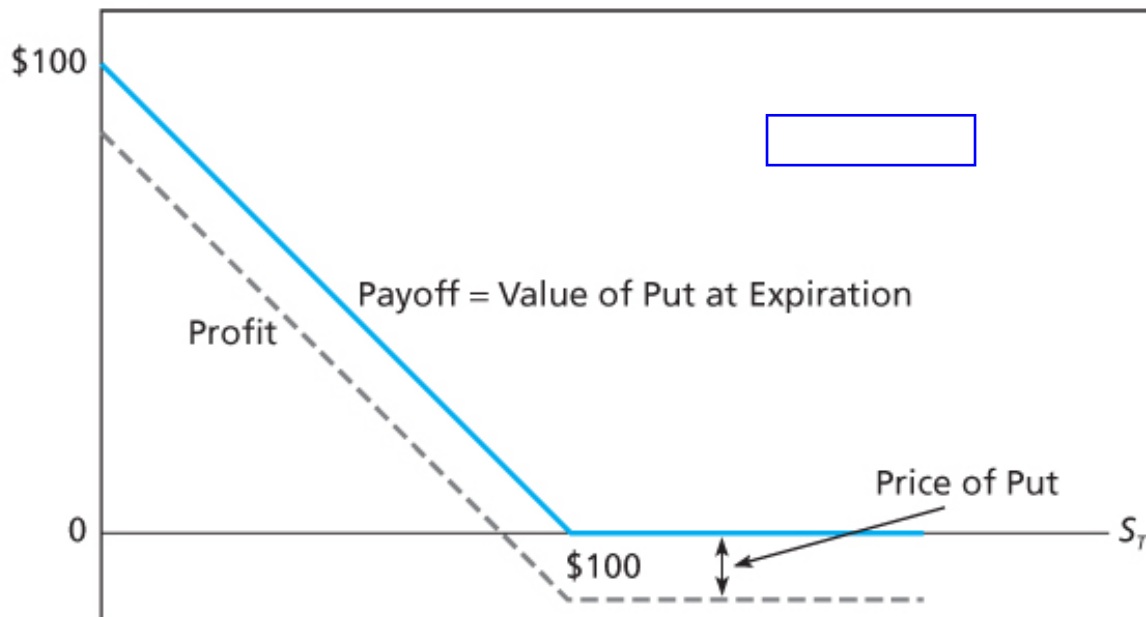
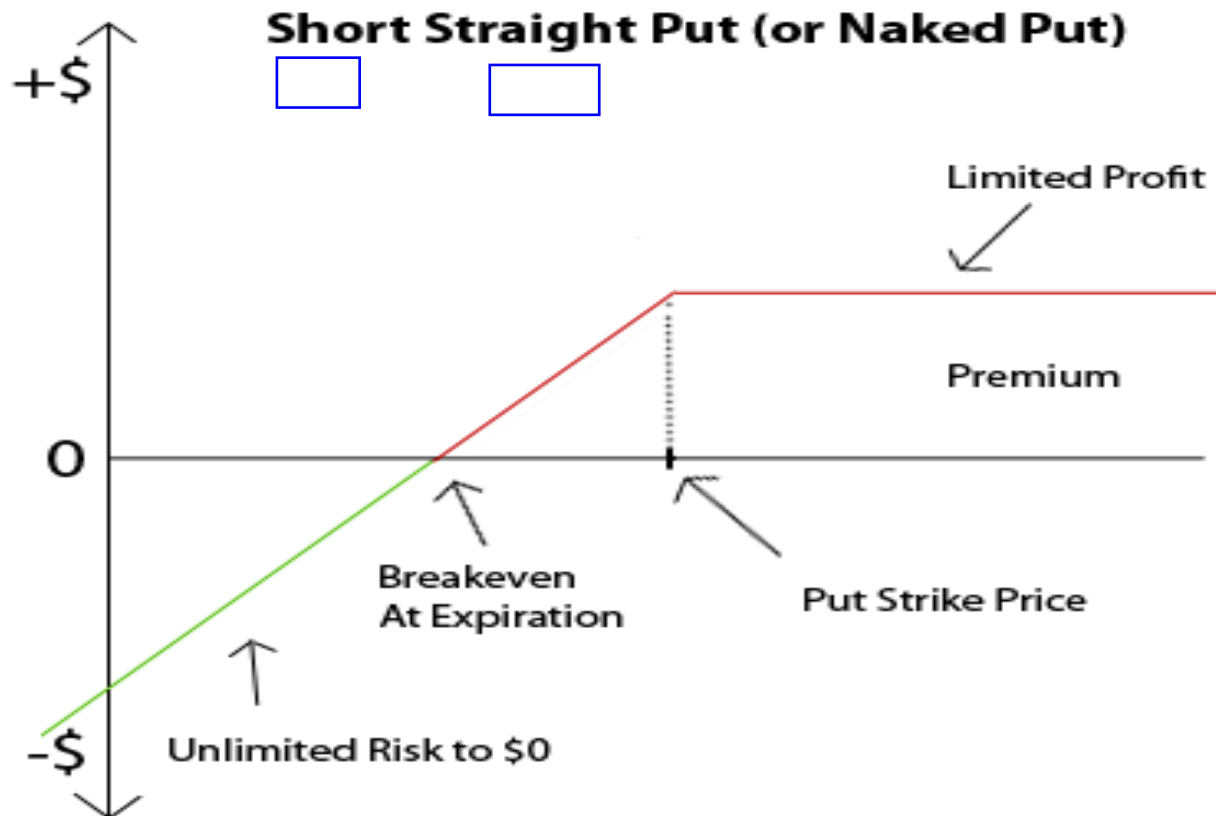


Figure 20.4 Payoff and profit to put option at expiration

Payoff and Profit to Put Writers at Expiration



Option versus Stock Investments

vs ?

- Could a call option strategy be preferable to a direct stock purchase?
- Suppose you think a stock, currently selling for \$100, will appreciate. 1 100 ; ? ?
- A 6-month call costs \$10 (contract size is 100 shares).
- You have \$10,000 to invest.

Option versus Stock Investments

- Strategy A: Invest entirely in stock. Buy 100 shares, each selling for \$100.
- Strategy B: Invest entirely in at-the-money call options. Buy 1,000 calls, each selling for \$10. (This would require 10 contracts, each for 100 shares.)
- Strategy C: Purchase 100 call options for \$1,000. Invest your remaining \$9,000 in 6-month T-bills, to earn 3% interest. The bills will be worth \$9,270 at expiration.

Option versus Stock Investment

Investment	Strategy	Investment
Equity only	Buy stock @ 100 100 shares	\$10,000
Options only	Buy calls @ 10 1000 options	\$10,000
Leveraged equity	Buy calls @ 10 100 options Buy T-bills @ 3% Yield	\$1,000 \$9,000

Strategy Payoffs

Portfolio	Stock Price					
	\$95	\$100	\$105	\$110	\$115	\$120
Portfolio A: All stock	\$9,500	\$10,000	\$10,500	\$11,000	\$11,500	\$12,000
Portfolio B: All options	0	0	5,000	10,000	15,000	20,000
Portfolio C: Call plus bills	9,270	9,270	9,770	10,270	10,770	11,270

$100 \times 100 = 10,000$; $100,000 \times 5 = 500,000$

Portfolio	Stock Price					
	\$95	\$100	\$105	\$110	\$115	\$120
Portfolio A: All stock	-5.0%	0.0%	5.0%	10.0%	15.0%	20.0%
Portfolio B: All options	-100.0 (%)	-100.0	-50.0	0.0	50.0	100.0
Portfolio C: Call plus bills	-7.3 (%)	-7.3	-2.3	2.7	7.7	12.7

$100,000 \rightarrow 1000 \times 100 = 100,000$; $100,000$

Rate of Return to Three Strategies

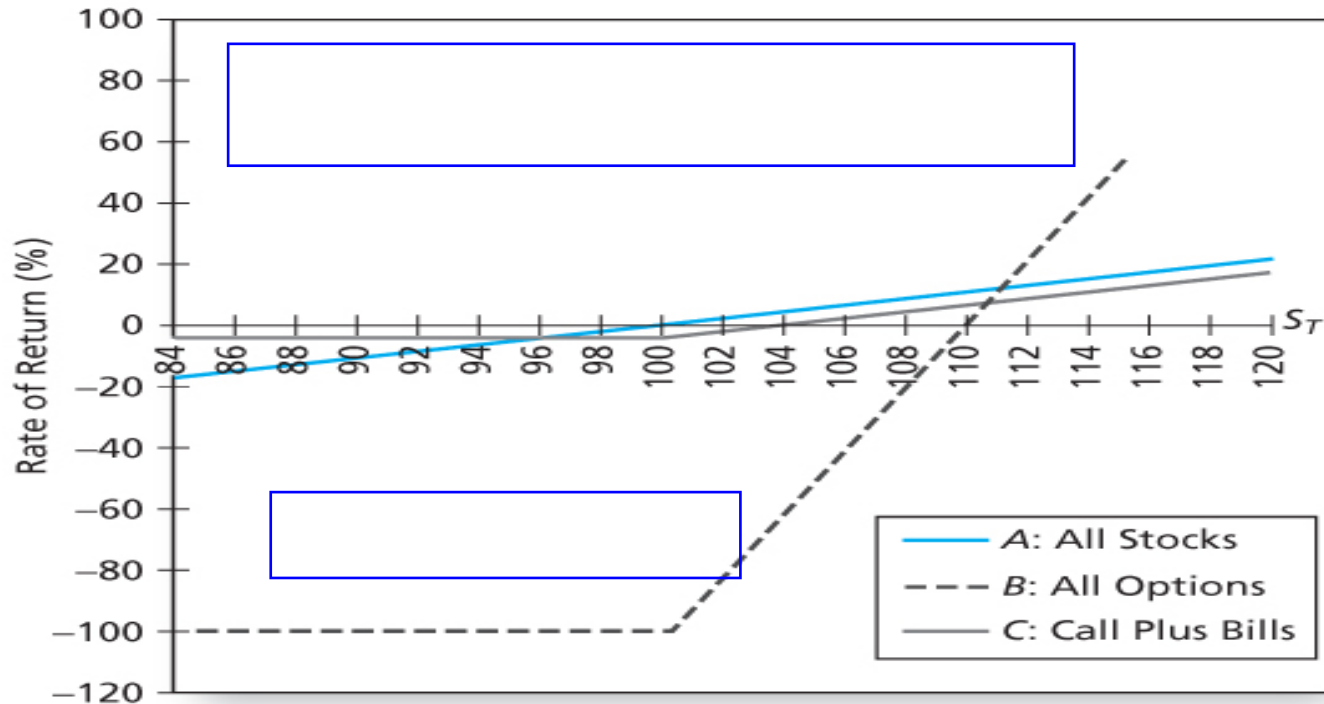


Figure 20.5 Rate of return to three strategies

Strategy Conclusions

- Figure 20.5 shows that the **all-option portfolio, B**, **responds more** than proportionately to changes in stock value; it is levered.
- Portfolio C, T-bills plus calls, shows the insurance value of options.
 - C 's T-bill position cannot be worth less than \$9270.
 - Some return potential is sacrificed to limit downside risk.